

W08 · M08 · 15 MINIT / MINUTES · Hari 2, 16:10 hingga 16:25

Hands-On SQA-AI Assistant Workshop

AI change request, impact analysis and CI review

Permohonan pindaan, analisis impak dan semakan CI dibantu AI

Raise a PPrISA10 change request for a new promo code, let the AI trace the impact through the RTM and test cases, and review the CI quality gate workflow.

opencode 1.18 · model Big Pickle (OpenCode Zen, free, no API key) · synthetic ShopFast data only

Bengkel SQA KRISAv2 dan PPrISA 2.0 · 14 hingga 15 September 2026 · Makmal Komputer Aras 11, Menara MyIPO

[Web guide \(copy prompts\)](#)

[Module hub](#)

[Example outputs \(GitHub\)](#)

Before you start / Sebelum mula

YOU WILL / ANDA AKAN

1. Run /change-request for promo code RAYA26
2. Run /impact-analysis (Plan agent)
3. Ask the Plan agent to review the GitHub Actions workflow
4. Agree the minimal regression set and rollback trigger

1. Jalankan /change-request untuk kod promo RAYA26
2. Jalankan /impact-analysis (ejen Plan)
3. Minta ejen Plan menyemak aliran kerja GitHub Actions
4. Setuju set regresi minimum dan pencetus rollback

CHECK FIRST / SEMAK DAHULU

- PowerShell is in C:\Users\<you>\sqa-krisa-bengkel\lab
- opencode starts and shows Big Pickle, OpenCode Zen (W00)
- Outputs from earlier workshops exist in outputs\ (or copy them from example-output)

Data rules / Peraturan data

- Synthetic ShopFast data only. Never paste real agency documents or personal data.
- Free models run outside Malaysia and may learn from prompts.
- AI output is a draft. A person checks and signs off.

OUTPUT

outputs/m08-change-request.md, outputs/m08-impact-analysis.md

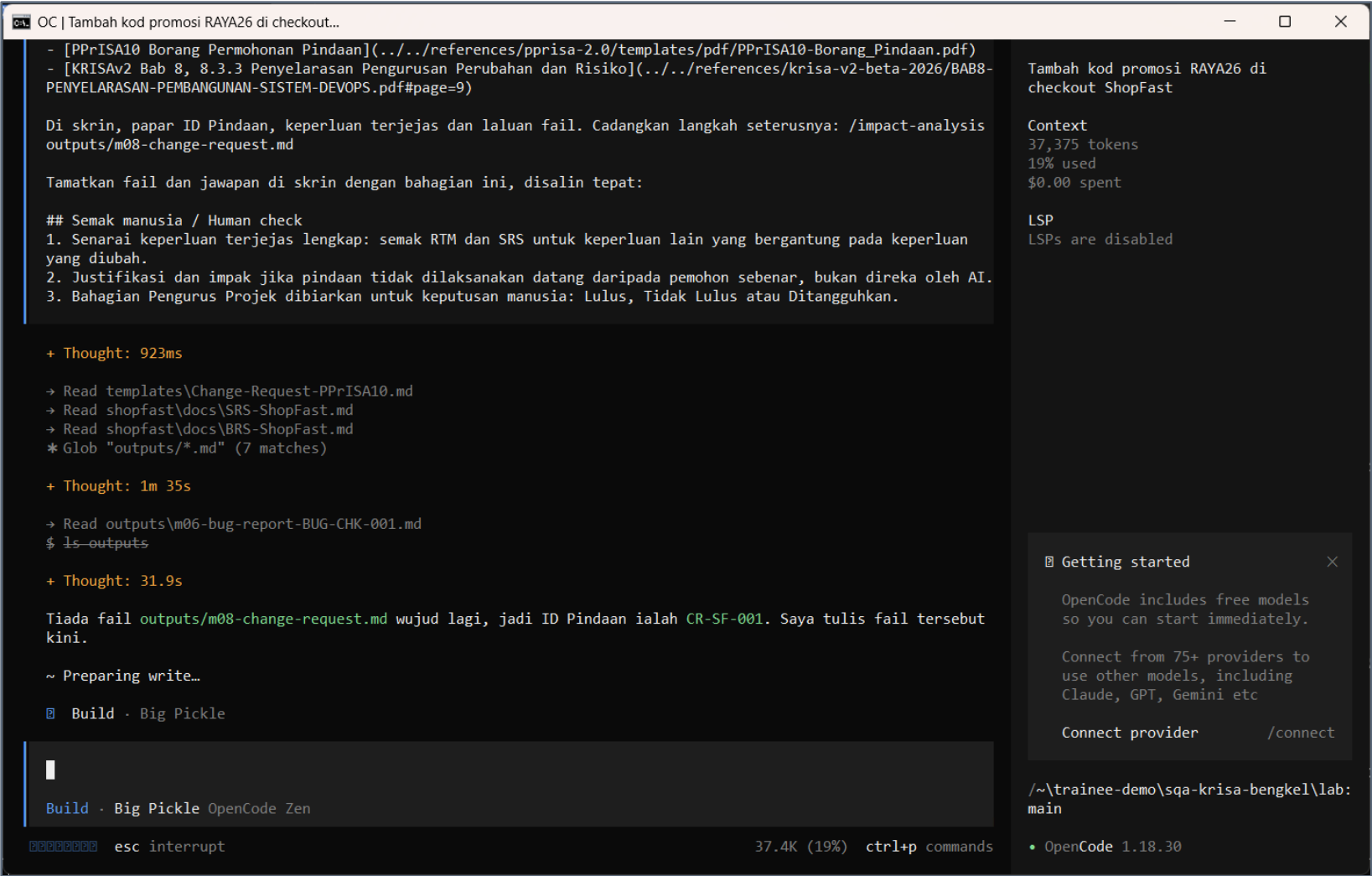
W08 · STEP 1 OF 4 · LANGKAH 1

Run /change-request for promo code RAYA26. The AI reads the documents first and may pause at Preparing write. When the Permission required dialog appears, read the preview and choose Allow once.

Jalankan /change-request untuk kod promo RAYA26. AI membaca dokumen dahulu dan mungkin berhenti seketika pada Preparing write. Apabila dialog Permission required muncul, baca pratonton dan pilih Allow once.

TYPE IN OPENCODE (BUILD)

/change-request Tambah kod promo RAYA26 (diskaun 26 peratus, sah 1 hingga 31 Mac 2027) pada checkout ShopFast

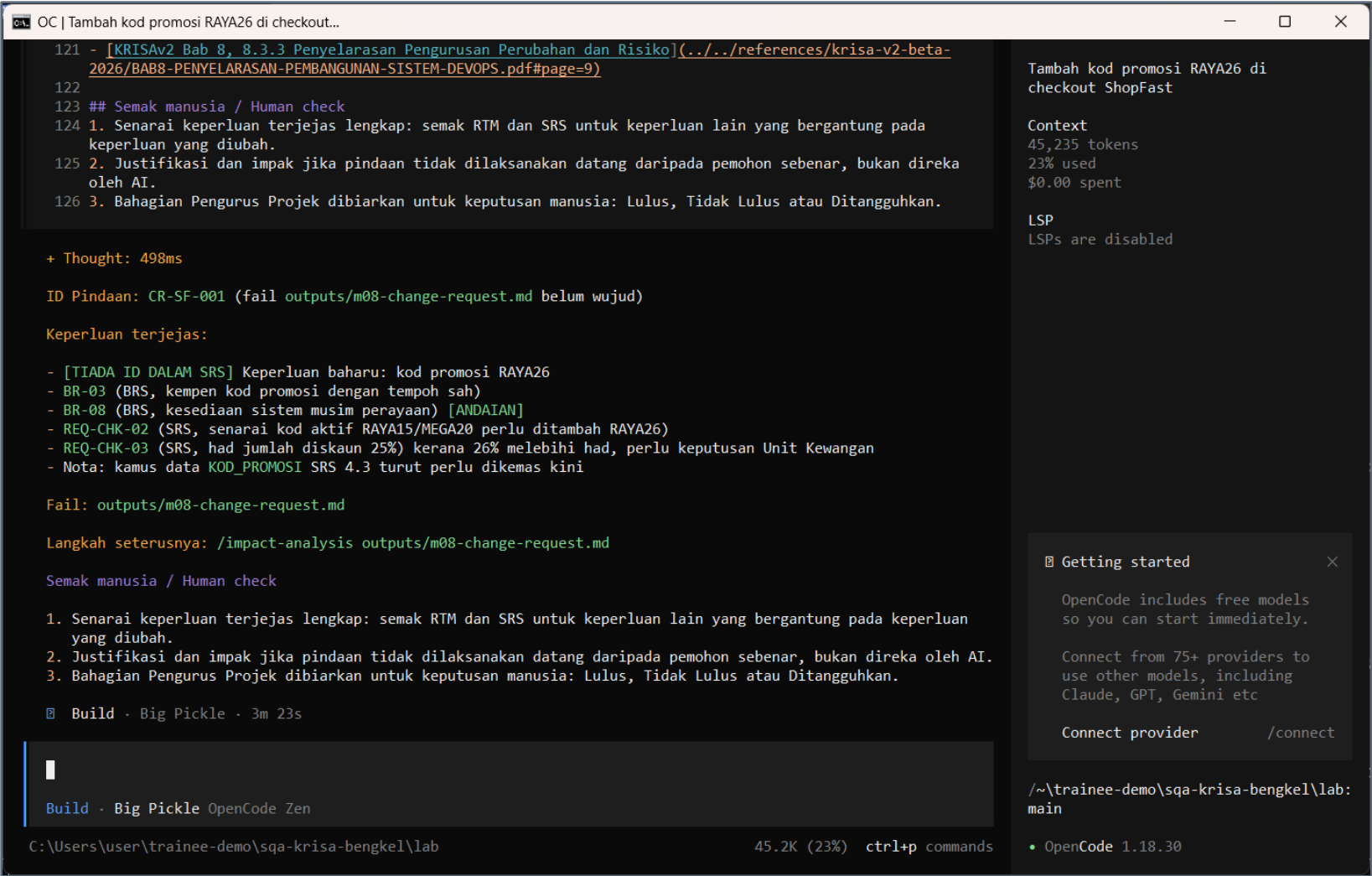


Allow once. The PPrISA10-style change request is saved.

Allow once. Permohonan pindaan gaya PPrISA10 disimpan.

TYPE IN OPENCODE

Allow once (Enter)



Run /impact-analysis in Plan mode.

Jalankan /impact-analysis dalam mod Plan.

TYPE IN OPENCODE (PLAN)

```
/impact-analysis RAYA26
```

OC | Analisis impak pindaan RAYA26

4	Sahkan RAYA26 tiada dalam senarai kod aktif (konfigurasi dipulihkan kepada RAYA15, MEGA20)	Semak fail konfigurasi kod promosi; pulihkan jika perlu	5 minit
5	Mulakan semula aplikasi	npm start	2 minit
6	Pengesahan selepas undur	Invoke-RestMethod http://localhost:3000/api/health; npm run test:api	10 minit

Pencetus, langkah dan RTO mengikut Rollback-Plan-template.md seksyen 2 dan 4. Pelan perlu diuji di staging sebelum pemasangan sebenar.

Rujukan

- PPrISA 2.0, 5.4.2 Pemantauan (impak pindaan): ../references/pprisa-2.0/PPrISA 2.0 Versi Beta Februari 2025.pdf#page=92 (../references/pprisa-2.0/PPrISA 2.0 Versi Beta Februari 2025.pdf#page=92)
- PPrISA 2.0, 4.2.3 c) Pengurusan Pindaan: ../references/pprisa-2.0/PPrISA 2.0 Versi Beta Februari 2025.pdf#page=73 (../references/pprisa-2.0/PPrISA 2.0 Versi Beta Februari 2025.pdf#page=73)
- KRISAv2 Bab 2, 2.11 Traceability Matrix: ../references/krisa-v2-beta-2026/BAB2-FASA-PERMULAAN.pdf#page=56 (../references/krisa-v2-beta-2026/BAB2-FASA-PERMULAAN.pdf#page=56)
- KRISAv2 Bab 8, 8.3.3 Penyelarasan Pengurusan Perubahan dan Risiko: ../references/krisa-v2-beta-2026/BAB8-PENYELARASAN-PEMBANGUNAN-SISTEM-DEVOPS.pdf#page=9 (../references/krisa-v2-beta-2026/BAB8-PENYELARASAN-PEMBANGUNAN-SISTEM-DEVOPS.pdf#page=9)

Untuk simpan, salin ke outputs/m08-impact-analysis.md dan tukar pautan kepada ../../references/...

Semak manusia / Human check

1. Semak RTM dan seni bina secara manual untuk kesan tidak langsung (endpoint lain, ujian, manual pengguna) yang tidak disenaraikan.
2. Anggaran tempoh dan kos disahkan oleh pasukan pembangunan, dan semua ANDAIAN diselesaikan sebelum keputusan pindaan.
3. Pencetus dan langkah undur boleh dilaksanakan dalam masa sasaran pemulihan dan telah diuji di staging.

Plan · Big Pickle · 2m 37s

Plan · Big Pickle OpenCode Zen

C:\Users\user\trainee-demo\sqa-krisa-bengkel\lab54.5K (27%)ctrl+p commands

Analisis impak pindaan RAYA26

Context54,530 tokens27% used\$0.00 spent

LSP
LSPs are disabled

Getting started
OpenCode includes free models so you can start immediately.
Connect from 75+ providers to use other models, including Claude, GPT, Gemini etc
Connect provider /connect

/~\trainee-demo\sqa-krisa-bengkel\lab:main

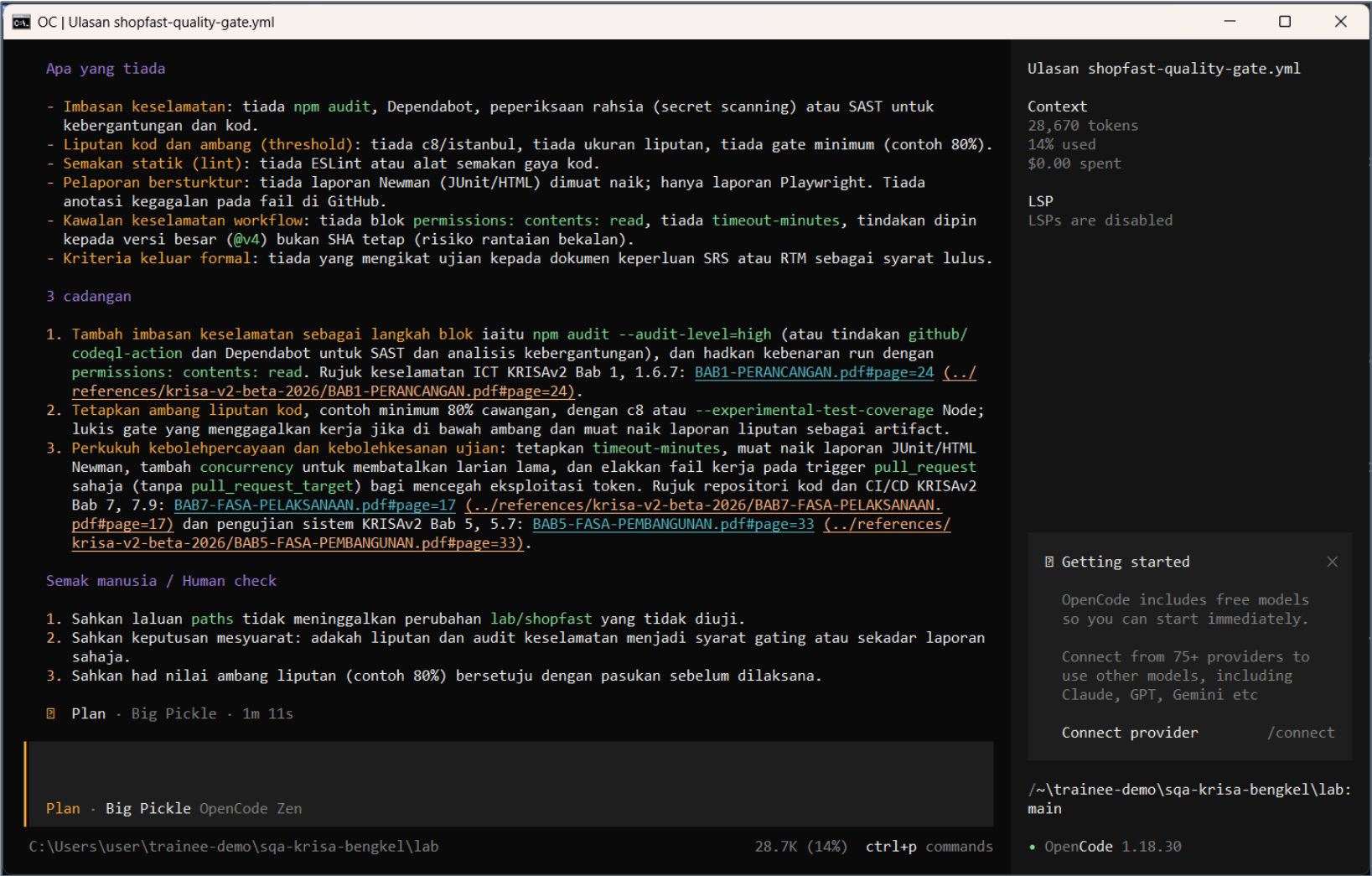
• OpenCode 1.18.30

Ask the Plan agent to review the GitHub Actions quality gate.

Minta ejen Plan menyemak quality gate GitHub Actions.

TYPE IN OPENCODE (PLAN)

Semak ../.github/workflows/shopfast-quality-gate.yml sebagai quality gate. Nyatakan apa yang disemak, apa yang tiada (contoh imbasan keselamatan atau ambang liputan) dan 3 cadangan. Jangan tulis fail.



Human check before you accept the output / Semakan manusia

RAYA26 at 26 percent breaks the 25 percent cap in REQ-CHK-03: did the AI notice?

RAYA26 pada 26 peratus melanggar had 25 peratus dalam REQ-CHK-03: adakah AI perasan?

The regression set is minimal and traced to REQ-CHK IDs

Set regresi adalah minimum dan dijejak ke ID REQ-CHK

Rollback has a clear trigger and an owner

Rollback mempunyai pencetus yang jelas dan pemilik

The AI draft becomes evidence only after this check. Record who checked it and when (Module 7 audit trail).

If something goes wrong / Jika ada masalah

opencode is not recognised

Close and reopen PowerShell so PATH refreshes. Still failing: `powershell -ExecutionPolicy Bypass -File .\setup\install-opencode.ps1 -Portable`

Scripts are disabled on this system

`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or run `opencode.cmd` instead of `opencode`

The model is slow or stops

Wait up to 2 minutes. Press Esc to interrupt, then send the prompt again. Try another free model with `/models`

Permission required dialog

Read the preview. Choose Allow once for files in `outputs/`. Reject anything outside the lab folder

File or command not found

Check the status bar path ends with `\lab`. Commands and `@files` are relative to the lab folder

Answer uses Indonesian words or dashes

Reply: Ikut AGENTS.md peraturan 1 dan 14, tulis semula dalam Bahasa Melayu Malaysia

Network or proxy error

`$env:HTTPS_PROXY = "http://proxy:port"` and `$env:NO_PROXY = "localhost,127.0.0.1"` before starting opencode

Port 3000 already in use

`$env:PORT = 3001` before npm start, and use `http://localhost:3001`

Abbreviations / Singkatan (1 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
AI	Artificial Intelligence	Software that can answer questions and write text or code by learning patterns from large amounts of data. <i>Kecerdasan Buatan. Perisian yang boleh menjawab soalan dan menulis teks atau kod berdasarkan pola daripada data yang banyak.</i>
API	Application Programming Interface	A doorway that lets one program ask another program for data or an action, without using a screen. <i>Pintu yang membolehkan satu program meminta data atau tindakan daripada program lain tanpa melalui skrin.</i>
CI	Continuous Integration, or Configuration Item	Two meanings. In DevOps, every code change is built and tested automatically. In configuration management, it is any controlled item such as a document or code file. <i>Dua maksud. Dalam DevOps, setiap perubahan kod dibina dan diuji secara automatik. Dalam pengurusan konfigurasi, ia ialah item terkawal seperti dokumen atau fail kod.</i>
CI/CD	Continuous Integration and Continuous Delivery	An automated pipeline that builds, tests and prepares every change for release. <i>Saluran automatik yang membina, menguji dan menyediakan setiap perubahan untuk dikeluarkan.</i>
ID	Identifier	A unique label, such as REQ-CHK-01, so that each item can be traced. <i>Pengecam, iaitu label unik seperti REQ-CHK-01 supaya setiap item boleh dijejak.</i>
KRISA	Kejuruteraan Sistem Aplikasi Sektor Awam	The Malaysian public sector guide for engineering application systems. KRISAv2 is the 2026 beta edition. <i>Buku Panduan Kejuruteraan Sistem Aplikasi Sektor Awam. KRISAv2 ialah edisi beta 2026.</i>

Abbreviations / Singkatan (2 of 2)

Short forms used in this workshop, in plain words. / Singkatan dalam bengkel ini, dengan bahasa mudah.

Abbreviation / Singkatan	Stands for / Maksud penuh	In plain words / Maksud mudah
MyIPO	Perbadanan Harta Intelek Malaysia	Intellectual Property Corporation of Malaysia, the host of this training venue. <i>Perbadanan Harta Intelek Malaysia, tuan rumah lokasi latihan ini.</i>
PPriSA	Panduan Pengurusan Projek ICT Sektor Awam	The Malaysian public sector guide for managing ICT projects. Codes such as PPriSA05 are its templates. <i>Panduan Pengurusan Projek ICT Sektor Awam. Kod seperti PPriSA05 ialah templatnya.</i>
REQ	Requirement	Prefix in requirement IDs such as REQ-CHK-01. CHK stands for checkout. <i>Keperluan. Awalan dalam ID keperluan seperti REQ-CHK-01. CHK bermaksud checkout.</i>
RTM	Requirements Traceability Matrix	A table linking each requirement to its design, tests and defects, so nothing is missed. <i>Matriks Kebolehkesanan Keperluan. Jadual yang memautkan setiap keperluan kepada reka bentuk, ujian dan kecacatan supaya tiada yang tertinggal.</i>
SQA	Software Quality Assurance	All the planned work that gives confidence the software meets its quality requirements. <i>Jaminan Kualiti Perisian, iaitu semua kerja terancang yang memberi keyakinan bahawa perisian memenuhi keperluan kualiti.</i>
TUI	Text User Interface	A full-screen program that runs inside a terminal window, like opencode. <i>Antara muka pengguna berasaskan teks, iaitu program skrin penuh dalam tetingkap terminal seperti opencode.</i>

Links / Pautan

[Web guide with copyable prompts](#)

[Module hub \(slides, references, templates\)](#)

[Example outputs from the real run](#)

[Lab README and command list](#)

[AGENTS.md rules used by the assistant](#)

[opencode documentation](#)

[opencode TUI commands and keys](#)

[OpenCode Zen free models and data policy](#)

[Bab 8, Jadual 8.1 Pemetaan Aktiviti KRISA ke DevOps \(p.6\)](#)

[Bab 8, 8.3.3 Penyelarasan Pengurusan Perubahan dan Risiko \(p.9\)](#)

[Bab 7, 7.9 Repositori dan Serahan Kod Sumber \(CI/CD\) \(p.17\)](#)